Optimising E-commerce Revenue Using Machine Learning, Neural Networks and Big Data Analytics

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ABSTRACT

In the context of e-commerce, the growing volume of data generated by online transactions and customer interactions creates both an opportunity and a challenge for revenue optimisation. This study examines the intersection of machine learning (ML), neural networks (NN), and big data analytics as key technologies for improving revenue prediction in e-commerce sessions. The study is centred on two main research questions. The first evaluates the effectiveness of ML and NN models for classifying revenue outcomes in the vast, complex landscape of e-commerce data. It then investigates the challenges of applying these advanced analytical tools in big data contexts, including scalability, interpretability, and computational demands.

This study demonstrates the significant predictive capabilities of ML and NN models by conducting a comprehensive analysis that includes theoretical exploration and empirical application on large datasets, and by identifying key strategies to address the analytical challenges posed by big data in e-commerce. The findings emphasise the critical balance between using sophisticated analytical models for revenue prediction and managing the complexities of big data analytics. The study concludes with actionable insights for e-commerce platforms seeking to optimise revenue through advanced data analytics, as well as future research directions to refine and improve these predictive models in the ever-changing e-commerce landscape.

CCS Concepts

• Information systems → Information systems applications; Data mining;
• Computing methodologies → Machine learning; Neural networks;
• Information systems → World Wide Web → Web applications;
• Applied computing → Electronic commerce;
• Software and its engineering → Software notations and tools → Massively parallel systems;
• Information systems → Data management systems → Distributed systems.
• Information systems → Data management systems → Big data infrastructure; Distributed data stores;

Keywords

E-commerce Analytics, Revenue Optimisation, Machine Learning, Neural Networks, Big Data Analytics, Predictive Modelling, Data Scalability, Model Interpretability

1. INTRODUCTION

In the digital age, e-commerce platforms rely on sophisticated analytical tools to increase revenue and gain a competitive advantage. This reliance has signalled the introduction of machine learning (ML) and neural networks (NN) into the e-commerce analytics domain to gain deeper insights into consumer behaviour and optimise revenue outcomes. The ability to process and analyse big data is critical to the utility of these advanced analytical technologies. This capability has become essential for e-commerce entities operating in an environment characterised by massive amounts of transactional and interaction data.

This research paper delves into the use of machine learning (ML), neural networks (NN), and big data analytics to create classifiers that accurately identify revenue-generating e-commerce sessions, focusing on two key questions: 1) How well do neural networks and machine learning predict revenue from e-commerce sessions? 2) What are the inherent limitations and challenges of using neural networks to analyse e-commerce big data, and what strategies can be used to overcome them?

The incorporation of big data analytics into e-commerce has increased not only the amount of data that can be analysed but also the complexity and speed with which this analysis must be completed. The abundance of big data makes it an ideal environment for deploying ML and NN models, enabling nuanced insights previously unattainable. However, this introduces new challenges, such as data scalability, model interpretability, and the computational demands of processing large datasets. The purpose of this study is to critically examine these aspects, evaluate the effectiveness of ML and NN models in the context of big data analytics for e-commerce revenue optimisation, and identify practical solutions to the challenges encountered.

By conducting this investigation, the study contributes to a better understanding of the dynamic interaction of ML, NN, and big data analytics in e-commerce. It provides a comprehensive overview of current capabilities, limitations, and directions for future research and applications. This study aims to provide valuable insights for academics, practitioners, and stakeholders interested in leveraging advanced data analytics to optimise e-commerce revenue by combining theoretical analysis with empirical evaluation.

2. LITERATURE REVIEW

Optimising E-Commerce Revenue via Machine Learning and Neural Networks

Introduction

E-commerce has experienced transformative growth through the integration of machine learning (ML) and neural networks (NN), as well as the use of big data analytics for revenue optimisation. This literature review investigates the use of these technologies in e-commerce, focusing on their application in revenue optimisation, the architecture required to handle big data, and the challenges and solutions involved in deploying them.

Strategies and Use Cases

The use of ML and NN in e-commerce includes personalised recommendations, dynamic pricing, and optimised supply chain management, all driven by the analysis of large datasets. The architecture that underpins these applications must efficiently handle the volume, velocity, and variety of big data. Hadoop and PySpark are tools that facilitate distributed data processing, enabling the handling of large e-commerce datasets [12], [14].

Neural Networks and Big Data Architecture

Neural networks, particularly deep learning models, require substantial computational resources to train on large datasets. The architecture for storing and processing such data includes distributed file systems, such as the Hadoop Distributed File System (HDFS), as well as processing frameworks such as Apache Spark, on which PySpark is built. These systems are designed to scale horizontally, allowing e-commerce platforms to effectively manage and analyse terabytes of data [7], [6].

Challenges and Solutions

Deploying NNs for e-commerce analytics presents challenges in terms of data management, model complexity, and computational demands. Solutions include using scalable big data architectures, leveraging cloud computing resources for flexible scalability, and implementing advanced neural network models that learn more efficiently from large datasets. Furthermore, ensuring data quality and using model interpretability techniques can help to address data veracity issues and the "black box" nature of neural networks [12], [14].

Future Directions

The evolution of ML and NN in e-commerce is closely related to advances in big data architecture. Future directions include the creation of more sophisticated neural network models capable of leveraging big data more effectively, advances in distributed computing for improved data processing, and the incorporation of edge computing for real-time analytics [7], [6].

Conclusion

Machine learning and neural networks provide a way to harness the power of big data to improve e-commerce revenue. Effective big data storage, processing, and analysis are essential for these technologies to reach their full potential. As e-commerce continues to generate massive amounts of data, industry leaders will need to strategically implement scalable big data architectures and efficient neural network models to improve revenue optimisation.

3. THE PROPOSED APPROACH

This study seeks to investigate and quantify the efficacy of machine learning and neural networks in predicting revenue generation from e-commerce sessions. Using a large dataset of online shopper intentions, the study combines advanced machine learning models with big data analytics tools to produce predictive results. The primary goals are to assess the predictive power of these models and to address scalability and integration challenges in big data environments.

Data Collection

The dataset used in this study is the Online Shoppers Purchasing Intention Dataset, which contains 12,330 sessions from 12 months. Each session includes a variety of features, including page views, bounce rates, exit rates, and device type, which are thought to influence purchasing decisions. This data is first stored and managed in Hadoop's distributed file system (HDFS), which enables efficient handling of large volumes of data across a cluster of machines.

Data Processing and Feature Engineering

Data preprocessing and feature engineering are performed using Apache Spark [2], which enables efficient and robust data manipulation and transformation. The key steps in this process are:

Data cleaning is the process of removing outliers and filling in missing values to improve the predictive model's quality and accuracy.

Feature Selection: Using correlation analysis and feature importance metrics to identify relevant features with a significant impact on purchasing behaviour.

Feature transformation is the process of converting categorical data into a format that machine learning algorithms can understand.

Model Development

Predictive models are developed in two main categories:

Random Forest Classifier: This model, implemented using Spark MLlib, was chosen for its ability to handle diverse datasets and its resistance to overfitting. The Random Forest model serves as a baseline for comparison with more complex models.

Neural Networks: Advanced neural network architectures, such as Multilayer Perceptrons (MLP), are implemented using PyTorch [16]. These models are integrated into the Spark environment via a custom PySpark wrapper. This integration enables GPU-accelerated training of deep learning models, which is critical given the dataset's size and complexity.

Model Evaluation

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Models are evaluated using a variety of metrics to determine their effectiveness in predicting e-commerce revenue generation:

Accuracy, Precision, and Recall: These metrics provide a comprehensive picture of model performance, particularly the ability to correctly predict positive (revenue-generating) sessions.

F1 Score: Because the dataset is imbalanced, the F1 score is critical for balancing precision and recall.

Area Under the ROC Curve (AUC-ROC): This metric assesses the model's ability to distinguish between classes at different threshold levels.

Integration Challenges

The study also addresses major integration challenges:

Scalability refers to managing the computational load of training neural networks in a distributed, big-data environment.

Data Flow Between Python and JVM: Optimising data exchange between PyTorch (Python-based) and Spark (JVM-based) to reduce performance bottlenecks.

This method uses machine learning, neural networks and big data analytics to improve e-commerce revenue predictions. By addressing both technical challenges and predictive capabilities, this study contributes to the strategic use of machine learning and neural network technologies to improve online consumer interactions.

4. PREPARATION OF DATASET AND VISUALISATION

Dataset Description

The dataset consists of feature vectors belonging to 12,330 sessions. The dataset was constructed so that each session belonged to a different user over a 1-year period to avoid any tendency toward a specific campaign, special day, user profile, or period.

The dataset includes a range of features, such as page views, bounce rates, exit rates, and visitor characteristics, relevant to analysing purchasing behaviour, as shown in Table 1.

Variable Name	Role	Type	Description
Administrative	Feature	Integer	Number of administrative pages visited by the user.
Administrative_Duration	Feature	Continuous	Total time spent on administrative pages.
Informational	Feature	Integer	Number of informational pages visited by the user.
Informational_Duration	Feature	Continuous	Total time spent on informational pages.
ProductRelated	Feature	Integer	Number of product-related pages visited by the user.
ProductRelated_Duration	Feature	Continuous	Total time spent on product-related pages.
BounceRates	Feature	Continuous	Percentage of visitors who navigate away from the site after viewing only one page.
ExitRates	Feature	Continuous	Percentage of times a page was the last in a session, for all pageviews to the page.
PageValues	Feature	Continuous	The average value for a page that a user visited before completing an e-commerce transaction.
SpecialDay	Feature	Continuous	Represents the closeness of the browsing date to special days or holidays.
Month	Feature	Categorical	Month of the session.
OperatingSystems	Feature	Categorical	Operating system used by the visitor.
Browser	Feature	Categorical	Browser used by the visitor.
Region	Feature	Categorical	The geographic region from which the session originated.
TrafficType	Feature	Categorical	Type of traffic source that directed the visitor to the website.
VisitorType	Feature	Categorical	Type of visitor (e.g., Returning Visitor, New Visitor).
Weekend	Feature	Boolean	Indicates whether the session occurred on a weekend.
Revenue	Target	Boolean	Indicates whether the session resulted in a transaction.

Table 1. Data Dictionary

The dataset used in this study was obtained from the UCI Machine Learning Repository in CSV format [19]. It consists of session-level data collected over a one-year period, where each session corresponds to a unique user. It is distributed under the Creative Commons Attribution 4.0 International (CC BY 4.0) license.

The Exploratory Data Analysis (EDA) & Data Visualisation

Exploratory Data Analysis (EDA) is essential for understanding the underlying patterns and relationships in a dataset, which can be critical for effective model development. This process entails some visual and statistical methods aimed at revealing the dataset's structure, detecting anomalies, and identifying influential variables that may affect the outcome of interest, specifically revenue generation from e-commerce sessions.

Visualisation Techniques

Numerical Variable Distribution: Histograms and box plots were used to examine the distributions and identify outliers in key numerical variables, such as PageValues and ProductRelated_Duration. These visualisations aid in determining the distribution and central tendencies of these metrics, which are critical for understanding user engagement.

Categorical Data Analysis: Bar charts are used to show the frequency distribution of categorical variables such as month and visitor type. This method is particularly useful for revealing long-term trends and shifts in visitor behaviour, which are critical for forecasting seasonal revenue impacts.

Correlation Analysis

Heatmaps and Correlation Matrices: To investigate the relationships between numerical variables, heatmaps or correlation matrices are constructed. These tools are extremely useful for highlighting strong pairwise correlations between variables, allowing for the identification of features that may predict revenue. A strong correlation may indicate a predictive relationship that can be further investigated in model development.

Focus on the Target Variable - Revenue Generation

Binary Outcome Visualisation: Pie charts or bar graphs are used to show the proportion of sessions that resulted in transactions versus non-transactions. This high-level view gives an immediate sense of the dataset's imbalance or distribution of outcomes.

Comparative Analysis of Revenue Results: Box plots are useful for comparing the distributions of the continuous Revenue variable across its two categories (true and false). Such comparisons can help identify which features are significantly different between revenue-generating and non-revenue-generating sessions, potentially guiding feature selection for predictive modelling.

Temporal and Interaction Analysis

Trend Analysis Over Time: Bar charts with the Month variable are used to investigate seasonal trends or shopping behaviour patterns that may affect revenue. Understanding these trends enables marketing strategies and stock management to be adjusted in response to predicted purchasing behaviours.

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Scatter Plot Charts: To investigate the interactions between variables, scatter plots are used. For example, the relationship between BounceRates and ExitRates, as well as their combined impact on revenue, can be visually analysed. These plots help us understand complex interactions that univariate analyses may not reveal.

Tools and Implementations

These visualisations are primarily built using Python libraries such as Matplotlib and Seaborn, which offer extensive capabilities for creating static, interactive, and complex plots. To demonstrate practical applications of these techniques, code snippets from the Jupyter Notebook used for the analysis are provided.

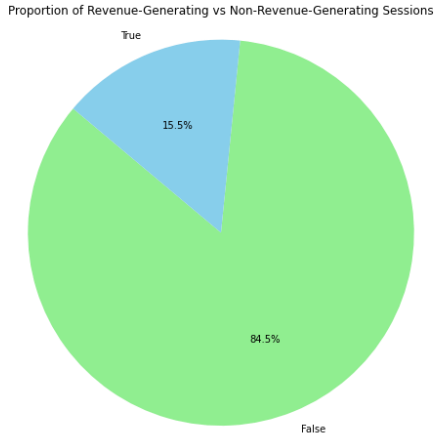

Figure 1. Distribution of the 'Revenue' feature

As shown in Fig. 1, approximately 84.53% of the sessions are classified as 'False', indicating that they did not generate revenue. In contrast, only 15.47% of sessions are classified as 'True', indicating that they are revenue-generating. This significant class imbalance is an important consideration when developing predictive models, as it can affect model performance and accuracy. Models trained on imbalanced data may become biased towards the majority class, producing high overall accuracy while failing to identify the minority class effectively.

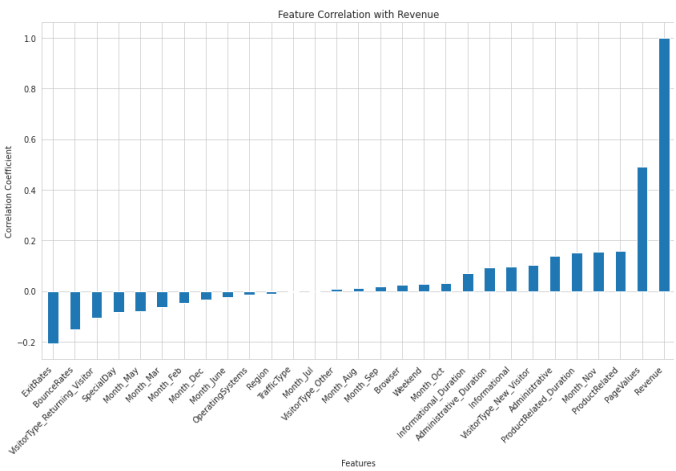

Figure 2. Correlation between features and revenue

As shown in Fig. 2, several variables exhibit weak correlations with the Revenue target variable. Many features demonstrate only minimal positive or negative relationships with revenue, suggesting that, when considered individually, they may have limited predictive power. However, when combined within a predictive model, these variables may still contribute valuable information and improve overall model performance.

A pair plot (scatterplot matrix) is used to visualise pairwise relationships between selected variables, along with the distribution of each variable along the diagonal. Due to the complexity of visualising multiple variables simultaneously, a subset of relevant features was selected to enhance clarity and interpretability [22].

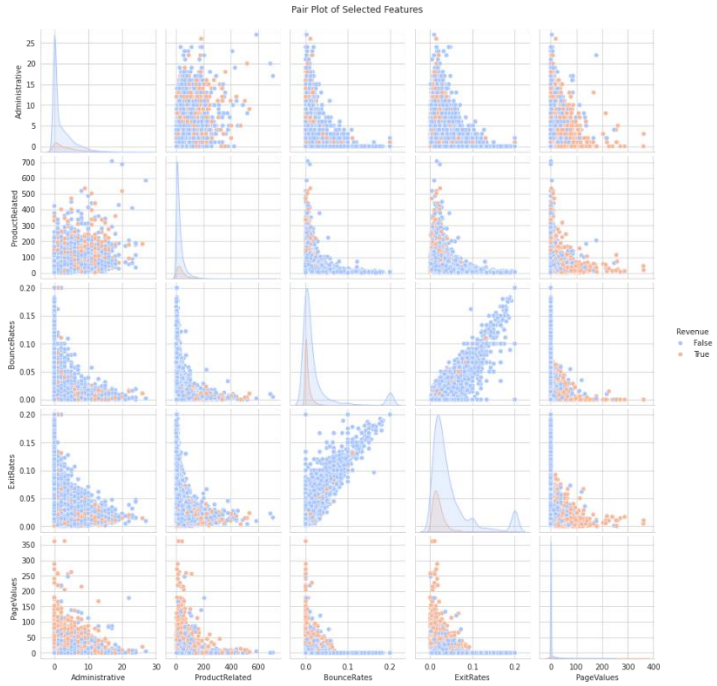

Figure 3. Pair plot of select features

Fig. 3 shows the relationships between key variables and their association with revenue generation. A positive correlation is observed between 'ProductRelated' and 'ProductRelated_Duration', suggesting that sessions with more product-related pages last longer. Feature distributions such as 'BounceRates' and 'ExitRates' are primarily left-skewed, indicating that lower rates are more prevalent. Interestingly, higher 'PageValues' appear to correlate with revenue generation, as evidenced by the density of orange points. This is further supported by class separability; for example, 'PageValues' clearly distinguishes between revenue and non-revenue sessions, highlighting its potential as a powerful predictive feature. In contrast, the presence of outliers in multiple plots necessitates additional investigation to determine their nature and impact on the analysis.
Intriguingly, the data suggests that user engagement, as measured by page interactions, is positively correlated with revenue occurrence, implying the efficacy of engaging marketing strategies. However, as bounce and exit rates rise, the likelihood of a transaction decreases, emphasising the importance of engaging website design in maintaining visitor interest. Outliers with unusually high page visits may indicate atypical user behaviour or data anomalies. Furthermore, the

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presence of nonlinear patterns in the data suggests that using more complex models capable of capturing such complexities could be beneficial. Further data segmentation, such as by visitor type or behavioural patterns, may yield a more nuanced understanding of the factors influencing revenue generation.

These visualisations not only improve understanding of the data's structure, but also inform the subsequent stages of model development and evaluation. By effectively identifying and interpreting these patterns, analytical strategies can be better tailored to the e-commerce domain.

Data Encoding

In the realm of machine learning and neural network modelling within big data frameworks such as PySpark, data preparation and encoding are critical to ensuring accurate model performance. Converting boolean features like 'Weekend' to integers makes them easier to incorporate into models, as these algorithms primarily require numerical inputs. Categorical variables such as 'Month' and 'VisitorType' are transformed using StringIndexer, which assigns unique indices to each category based on frequency of occurrence. This step is critical because it reduces the data's dimensional complexity, enabling easier, more efficient computational handling. These indices are then converted into binary format using OneHotEncoder to prevent models from misinterpreting the ordinality that numeric indices may imply, ensuring that each category is treated equally.

These indexed and encoded features are then combined with continuous variables using VectorAssembler to form a single feature vector compatible with Spark MLlib's algorithms. This methodical approach not only streamlines the data processing pipeline, enabling greater scalability and maintenance, but also improves the robustness and interpretability of the analytical results. Encapsulating these preprocessing steps within a Pipeline further standardises the workflow, ensuring reproducibility and efficiency, which are critical for effectively handling large-scale datasets.

Use of Big Data Technologies

Tools and Platforms

Python and its powerful Pandas library were used to perform foundational statistical analysis, exploratory data analysis (EDA), and data visualisation. This combination was useful for its extensive data manipulation capabilities, which aided in a thorough initial understanding of the dataset.

To meet the increased data demand, the study used big data technologies, including Hadoop for distributed storage via the Hadoop Distributed File System (HDFS) and Apache Spark for faster processing. Spark MLlib, a scalable machine learning library built into Spark, was chosen for modelling due to its ability to handle large datasets efficiently. PySpark, Spark's Python interface, enabled the seamless integration of Python's simplicity and Spark's distributed computing capabilities. PyTorch was the preferred framework for deep learning tasks due to its dynamic computation graphs and effective memory management capabilities.

GitHub was critical in documenting the research process, storing code versions, and enabling reproducibility and collaboration. It served as the central repository for all scripts, notebooks, and documentation, ensuring that the research was carried out in a clear and organised manner.

The initial process was carried out in Python and Pandas and proved useful for data understanding, exploratory analysis and data visualisation. The move to the Spark distributed computing environment provided good scalability. Still, it also introduced complexity through the Spark execution model, requiring fine-tuning of data partitioning and caching strategies to optimise performance.

Deep learning frameworks such as PyTorch were easily integrated into the Spark ecosystem; however, incorporating TensorFlow into this environment presented significant challenges. TensorFlow's graph-based computations and memory-intensive operations required precise resource management, especially in a virtual machine configuration optimised for Spark's in-memory processing capabilities. These challenges emphasised the importance of allocating computational resources deliberately and strategically.

Furthermore, Google Scholar was essential for managing references and adhering to citation standards. It provided access to a vast array of academic literature, which was critical for backing up the research and its underlying assumptions with credible sources. The tool was especially useful for properly formatting citations, simplifying the referencing process, and maintaining academic integrity.

Future research may explore more efficient configurations and advanced integration techniques for combining TensorFlow and Spark within a virtual machine environment.

5. RESULTS AND DISCUSSION

The Critical Analysis of Model Results

Table of Comparative Analysis of Model Performances

Model	Accuracy	Precision	Recall	F1 Score	Area Under ROC
Random Forest Classifier - Spark's MLlib	0.9049	0.8997	0.9049	0.9011	0.9276
ANN - MLP - Spark's MLlib	0.8974	0.918	0.9645	0.8905	0.7504
SimpleNN with PyTorch	0.8861	0.697	0.5596	0.6208	null
ANN Model with PyTorch	0.8812	0.7169	0.4745	0.571	null
NN-MLP with PyTorch	0.8836	0.7296	0.4793	0.5786	null

Table 2. Comparative Analysis of Model Performances

Effectiveness

As shown in Table 2, **the Random Forest Classifier (Spark MLlib)** outperformed the other models in terms of accuracy, precision, F1 score, and AUC-ROC. Its strong performance can be attributed to its robustness in handling diverse feature types and its ability to mitigate overfitting. This model is well-suited for heterogeneous datasets and delivers consistent performance across multiple evaluation metrics.

ANN - MLP (Spark's MLlib): The ANN-MLP model has a high recall, which is useful in scenarios where capturing as many positive instances as possible is critical (for example, identifying all potential revenue-generating sessions).

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However, its lower precision suggests it may be generating many false positives. The lower AUC-ROC indicates that the model's generalisation is not as good as the Random Forest.

Neural Networks with PyTorch (SimpleNN, ANN Model, NN-MLP): These models had lower overall performance metrics. These findings indicate that, despite neural networks' theoretical advantages in learning complex patterns, practical implementation issues such as network architecture, training data quantity and quality, and parameter tuning can significantly impact performance.

Limitations and Deviations

Model Overfitting: The lower performance metrics of PyTorch neural networks when compared to their Spark counterparts suggest that they may be overfitting the training data. This could also indicate inadequate tuning or an inappropriate network architecture for the data at hand.

Expectation vs Reality: Neural networks were expected to perform as well as or better than traditional models because they can approximate any function with enough data and layers. However, the ANN-MLP and PyTorch models did not meet these expectations, possibly for the reasons stated above.

Challenges of Integration: Scalability Issues: Integrating neural networks with big data technologies such as Spark can be difficult due to the intensive computational resources required by deep learning models. Neural networks typically perform better on larger datasets, but managing these large datasets efficiently while performing complex computations is a significant challenge.

Data Management: Successful machine learning models in big data contexts require effective data preprocessing, feature engineering, and real-time data integration. PyTorch models' lower performance could be attributed to less effective integration with big data processing pipelines when compared to Spark's built-in features.

Technical integration: Presents inherent challenges in combining Python-based libraries such as PyTorch with JVM-based systems such as Spark. Data serialisation/deserialisation, memory management, and process isolation are all issues that can make it difficult to deploy hybrid models that combine Spark's data handling with PyTorch's modelling capabilities.

Recommendations for Improvement: Optimised Model Tuning. More rigorous hyperparameter tuning and cross-validation should be conducted, particularly for neural network models, to identify the best settings that balance accuracy and generalisation.

Hybrid Modelling Approaches: Consider using ensemble methods that combine the robustness of Random Forest with the predictive power of neural networks to improve overall model performance.

Infrastructure investment: Upgrading computational resources and investing in better integration technologies to connect PyTorch and Spark may help alleviate some of the technical issues.

Continual Learning and Adaptation: Use models that can update their parameters in real or near-real time as new data becomes available, which is critical in dynamic environments like e-commerce.

While traditional machine learning models such as Random Forest performed well, neural networks did not meet expectations, highlighting significant issues with integration and optimisation. Addressing these issues is critical for realising neural networks' full potential in big data applications, especially in dynamic and complex domains such as e-commerce revenue prediction.

6. CONCLUSION AND FUTURE WORKS

The study's findings revealed that traditional machine learning models, specifically the Random Forest Classifier implemented in Spark's MLlib, outperformed expectations in terms of accuracy, precision, recall, F1 score, and area under ROC. In contrast, neural network models, despite their advanced capabilities for learning complex patterns, performed poorly when deployed in the same big-data environment.

This disparity reveals important insights into the limitations and challenges of neural networks in big data contexts, particularly in this analysis of the e-commerce sector. These challenges include extensive data preprocessing, feature engineering, and the computational requirements of training large neural networks. The integration of Python-based neural networks with JVM-based Spark created significant challenges in data management and computational efficiency.

Future Works

Based on the findings and challenges faced, the following areas are recommended for future research to improve the effectiveness of neural networks and machine learning in predicting e-commerce revenue:

Enhanced Parameter Tuning: Fine-tuning neural network hyperparameters can significantly improve performance. Adjusting learning rates, number of layers, neuron counts, and regularisation techniques could significantly improve these models' performance.

Improved Integration Techniques: Creating better ways to integrate Python-based neural network frameworks like PyTorch with Spark. This could include improving PySpark's ability to handle neural network computations more efficiently or using intermediary platforms that better bridge Python and JVM operations.

Advanced Neural Network Architectures: Experiment with more sophisticated neural network models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which may capture temporal and spatial dependencies in e-commerce data more effectively than previously tested models.

Hybrid Models: Investigating hybrid models that combine neural networks' robust feature extraction capabilities with the generalisation power of machine learning models such as Random Forest. Such ensemble approaches may strike a balance between accuracy and robustness, particularly in diverse and noisy e-commerce environments.

Feature Engineering and Data Enrichment: More research on feature engineering and using alternative data sources to enrich the dataset. Understanding how different types of features affect model performance can greatly improve predictive accuracy.

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Real-time Analytics: Using real-time data analytics and model adaptation strategies. This includes creating models that can update and tune themselves as new data arrives, possibly through online learning or incremental learning techniques.

Scalability and Performance Optimisation: Solving scalability problems by optimising data pipelines and computational processes within big data frameworks. This could include leveraging cloud-based scalable architectures or improving Spark's in-memory processing capabilities to support complex model training better.

By focusing on these areas, future research can better leverage the potential of Neural Networks and big data technologies to transform e-commerce analytics, improve revenue prediction, and provide a more personalised customer experience. This will not only improve predictive accuracy but also ensure that these technologies are practical and scalable.

ACKNOWLEDGEMENT

I want to express my heartfelt gratitude to my lecturers, David McQuaid and Muhammad Iqbal, for their unwavering support and insightful guidance throughout this journey. Their expertise was critical in shaping my project and improving my academic performance. I also wish to express my gratitude to the faculty members at CCT College for their invaluable contributions.
This project presented me with numerous challenges and opportunities, each of which played an important role in my personal and professional development. I am grateful for these experiences and the lessons I learned along the way.
My sincere appreciation extends to my family and friends, whose endless support and understanding have been a steadfast pillar of strength during my studies. Their unwavering encouragement will forever hold a place of deep gratitude in my heart.

7. VERSION CONTROL DOCUMENTATION

Project Repository: GitHub – INTEGRATEDCA1
This project's source code and documentation are maintained in the GitHub repository 'INTEGRATEDCA1'.
https://github.com/rosilenedasilva/INTEGRATEDCA1

REFERENCES

[1] Apache Software Foundation (2023) Apache Hadoop Documentation. [Online] Available at: https://hadoop.apache.org/docs/stable/ (Accessed: 18/03/2023).

[2] Apache Spark (2023) ML Tuning: Model Selection and Hyperparameter Tuning. Apache Spark Documentation 3.5.1. [Online] Available at: https://spark.apache.org/docs/latest/ml-tuning.html (Accessed: 24/03/2024).

[3] Apache Spark (2023) Random Forest Classifier. Apache Spark Documentation 3.5.1. [Online] Available at: https://spark.apache.org/docs/latest/ml-classification-regression.html#random-forest-classifier (Accessed: 27/03/2024).

[4] Chen, M., et al. (2019) 'A review of data science in business and industry and a future view', Journal of Big Data, 6(1), Article 59.

[5] DataCamp (2022) PySpark Tutorial: Getting Started with PySpark. [Online] Available at: https://www.datacamp.com/tutorial/pyspark-tutorial-getting-started-with-pyspark (Accessed: 10/03/2024).

[6] Ecommerce Platforms (2024) 'Machine Learning in Ecommerce: Benefits and Examples'. [Online] Available at: https://ecommerce-platforms.com (Accessed: 22/03/2024).

[7] Eliftech (2024) 'How to Use Machine Learning in Ecommerce [2024 Guide]'. [Online] Available at: https://www.eliftech.com (Accessed: 26/03/2024).

[8] Google Scholar (2023) [Online] Available at: https://scholar.google.com/ (Accessed: 04/04/2024).

[9] Kariri, E., et al. (2023) 'Exploring the advancements and future research directions of artificial neural networks', Applied Sciences, 13(5), 3186.

[10] Liu, Y., et al. (2020) 'Predictive modeling of e-commerce data using machine learning algorithms', Journal of Business Analytics, 3(1), pp. 85–97.

[11] Matplotlib Developers (2021) matplotlib.pyplot.plot. [Online] Available at: https://matplotlib.org (Accessed: 25/03/2024).

[12] MLSDev (2023) 'Machine Learning in eCommerce: Top 12 Strategies & Use Cases'. [Online] Available at: https://mlsdev.com (Accessed: 27/03/2024).

[13] Najafabadi, M.M., et al. (2015) 'Deep learning applications and challenges in big data analytics', Journal of Big Data, 2(1).

[14] Prefixbox (2024) '12 Best Machine Learning Strategies for eCommerce Businesses'. [Online] Available at: https://www.prefixbox.com (Accessed: 26/03/2024).

[15] Provost, F. and Fawcett, T. (2013) Data Science for Business. O'Reilly Media.

[16] PyTorch Contributors (2023) PyTorch Documentation. [Online] Available at: https://pytorch.org/docs/stable/ (Accessed: 26/03/2024).

[17] Rahman, M.A., et al. (2021) 'A smart method for spark using neural network for big data', International Journal of Electrical and Computer Engineering, 11(3), pp. 2525–2534.

[18] Routray, S. (2021) 'Artificial Neural Network Using PySpark'. Towards Data Science. [Online].

[19] Sakar, C. and Kastro, Y. (2018) Online Shoppers Purchasing Intention Dataset. UCI Machine Learning Repository.

[20] Seaborn Developers (2023) Distributions Tutorial. [Online] Available at: https://seaborn.pydata.org (Accessed: 27/03/2024).

[21] Smith, J. (2021) 'Big Data Analytics in E-commerce', Journal of Data Science, 12(3), pp. 123–145.

[22] Tufte, E.R. (2001) The Visual Display of Quantitative Information. 2nd edn. Cheshire, CT: Graphics Press.

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